

Exam in Topology Optimization Spring 2026

The exam consists of the following 6 tasks:

1. Find an interesting problem to solve using topology optimization. It must not be exactly the same as one we have already solved in class. To choose a problem, you can
 - a. Get inspiration from the exercises and lectures.
 - b. Get inspired by the literature.
 - c. Use a (simplified) version of a problem you work on in other courses/projects.
 - d. Use your imagination!

We can discuss your choice of problem in class or in the breaks, if you wish.

2. Implement a topology optimization solver for your problem using Python, FEniCS, pyadjoint, and the MMA code on Brightspace.

The solver you create **must** use robust optimization with double filtering (see lecture 10).

If you have 3 or more cores in your computer, then it is recommended, but not required, to parallelize the evaluations of the design cases (see lecture 8).

3. Use three design realization in the robust optimization with η -values given as

$$\eta_d = \eta_i - \Delta\eta, \quad \eta_i = 0.5, \quad \eta_e = \eta_i + \Delta\eta$$

Make a choice of the filter radius and $\Delta\eta$ such that you get working design.

4. Use two design realization in the robust optimization with η -values given as

$$\eta_d = 0.5 - \Delta\eta, \quad \eta_e = 0.5 + \Delta\eta$$

Use the same filter radius and $\Delta\eta$ which you used in part 3.

From the optimized design variables, calculate the “intermediate” design using $\eta = 0.5$ and the final value of β .

Calculate the cost function value of the intermediate design.

5. Compare the performance and geometries of the designs obtained in points 3 and 4. For the problem you have solved, is it a good idea to use only the eroded and dilated designs in the optimization (point 4) or is it better to include also the intermediate design in the optimization (point 3)?

6. Make an A3 poster presenting your problem, the topology optimization solution, and the numerical experiments. Explain the needed theory and techniques, such that the poster is understandable by your fellow students.

The poster must include

- a. The mathematical definition of the PDE(s) and boundary conditions used (in strong or weak form).
- b. The mathematical definition of the considered optimization problem.
- c. Explanation of the double filtered robust methods used in points 3 and 4.
- d. A comparison of the designs obtained and your answer to the question in point 5.

(assignment continues on the next page...)

You must hand in the following using WiseFlow before the deadline stated in WiseFlow:

1. The poster in PDF format.
2. Python code for the topology optimization solver you have implemented.
Include **all** python files needed to run your code, also files handed out on Brightspace, such as `mma.py`.
You can place all your python code in a zip file, if needed.

Note: This exam assignment is **individual**. You must hand in a unique product made solely by yourself.

Don't hand in other students' code! You can use and modify code handed out on Brightspace, if you make proper references. See the rules here:

<https://studerende.au.dk/en/examinations/cheating-at-exams/know-the-rules-when-you-have-an-exam>

You are *encouraged* to discuss the theory, techniques, code etc. with your fellow students, the teacher etc., but never share your code or use other people's code without proper reference.